

From simulation log, find which memory's GO-1D-REG is failing.

↳ in our eg:- for ram1.

Open waveform => vsim wave.wlf

Add the following signals:

MBISTPG-GO
BIST-DN
BIST-CMP } from controller
(in our case controller c1 as it is controlling ram1)

After BIST-DN has become 1, check where MBISTPG-GO is 0.

Go to the preceding pulse of BIST-CMP.
(because the preceding compare has only caused the MBISTPG-GO to become low)

For ram1, get the expected D/P and actual D/P

ram1-interface
↓
BIST-EXPECT-DATA

↓
ram1/q

(∵ we are having comparator inside the interface).

In our eg:- BIST-EXPECT-DATA ⇒ 2'h2
↳ the expected data is in compressed form

$$\begin{array}{cc} \Downarrow & \\ \frac{10\ 10\ 10\ 10}{a} & \frac{10\ 10}{a} \end{array}$$

$aa \Rightarrow$ original expected data

But actual data is $8'h\ 00$ as seen in the waveform.

You can load the memory's CLKWR, CLKR, AR, AW, WE, GWE, RE, D in the waveform.

$\rightarrow$ We can see that when

$$\begin{array}{l} WE = 1 \\ GWE = ff \end{array}$$

$$\Rightarrow AW = 0$$

$$\Rightarrow D = aa$$

after $WE = 0$

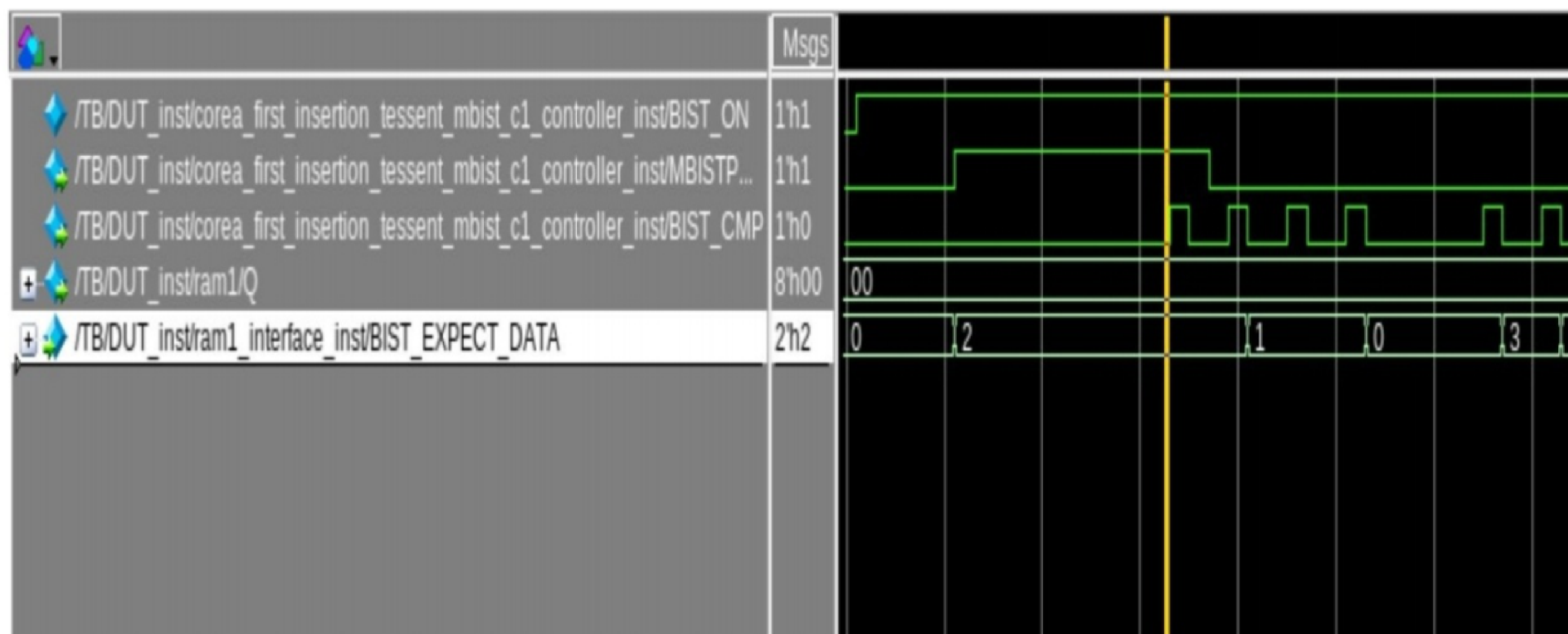
check $RE = 1$

$$\Rightarrow AR = 0$$

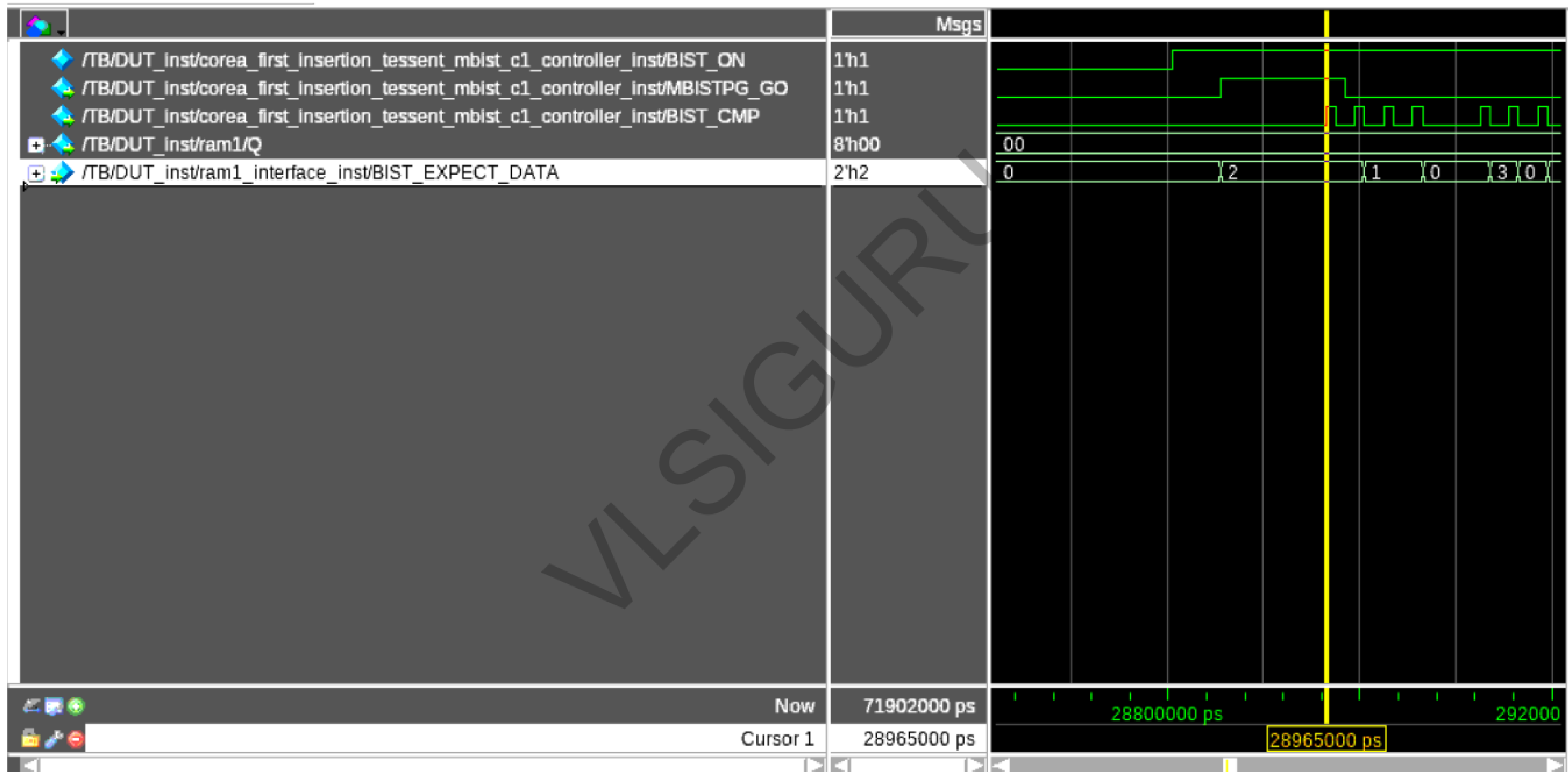
$$Q = 00.$$

check at BIST_CMP

In our case Q of mem is 0 as we have forced it to 0.



VLSIGURU



fn of the memory in our design (v file of memory)

posedge

clkW

↳ WE = 1

↳ GWE = 1

↳ Mem[AW] = D

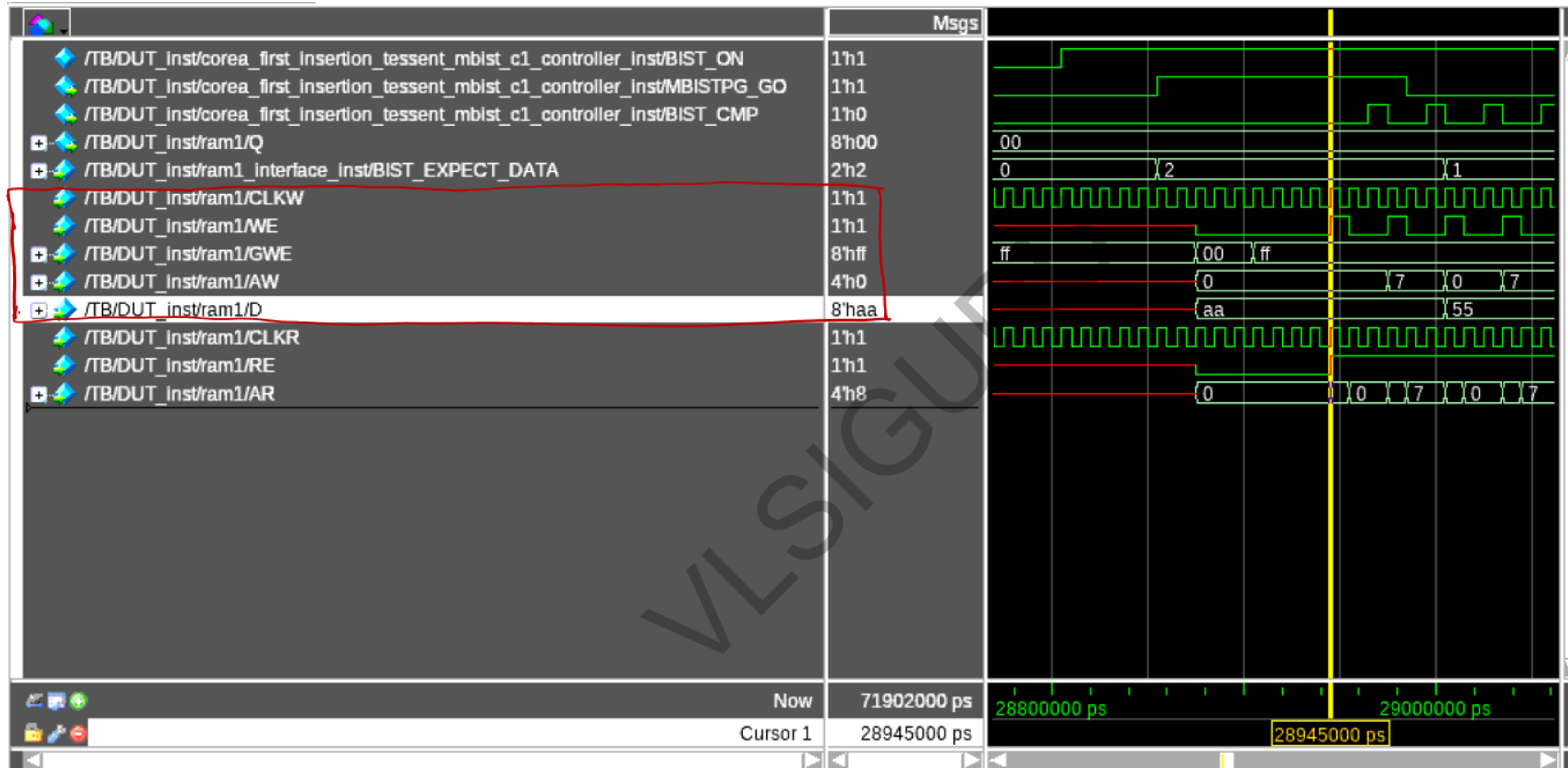
GWE = 0

Mem[AW] = Mem[AW]

posedge CLKR

↳ RE = 1

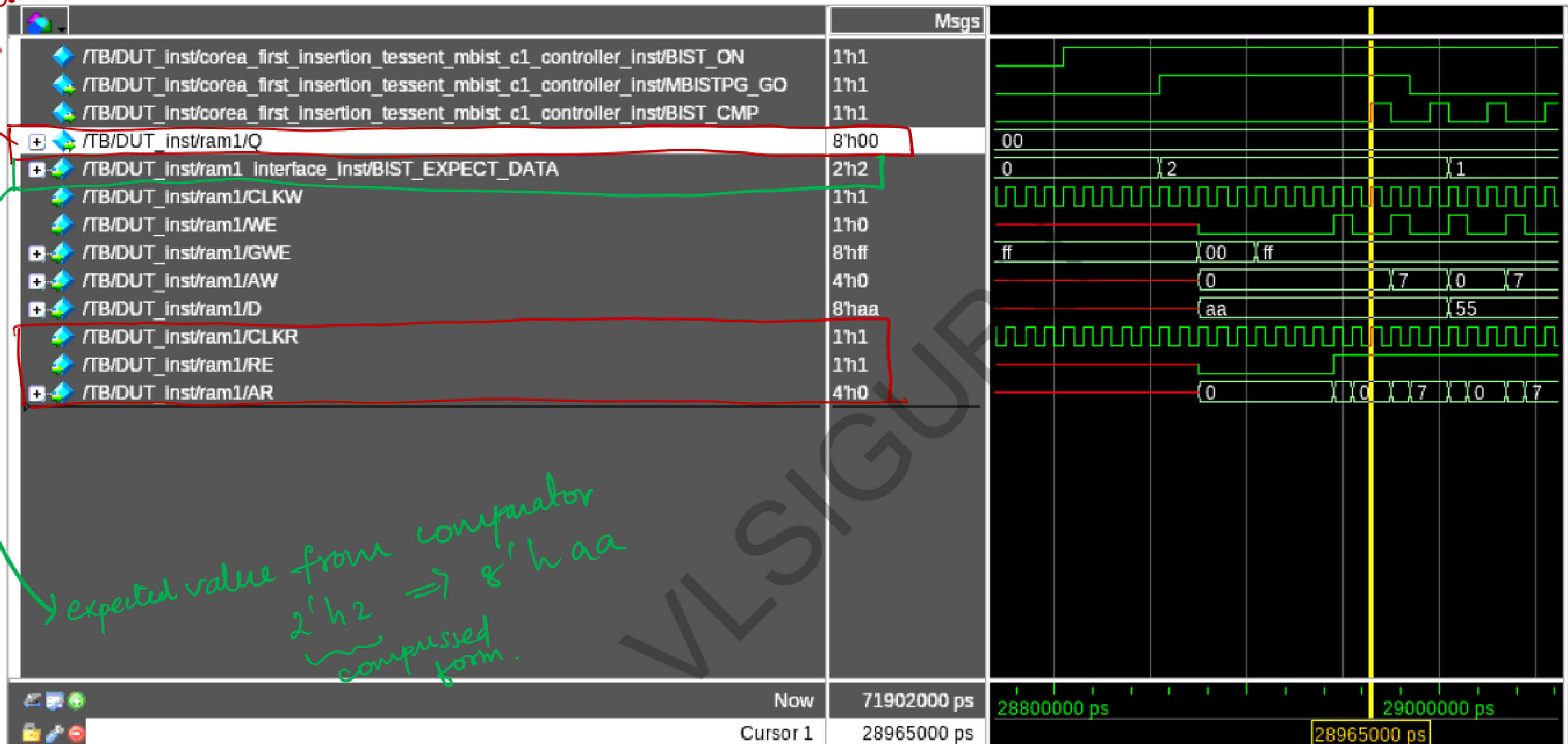
↳ Q = Mem[AR]



$AW = 0$
 $D = 8'haa$

posedge $CLKW$, $WE = 1$, $GWE = 1$
 $\hookrightarrow Mem[0] \leftarrow 8'haa$

ok from mem



CLKR = 1

RE = 1

AR = 0

Mem[AR] $\rightarrow$ Mem[0] $\Rightarrow$ 00